

Accessible Presentations with LuaLaTeX

A tagged-PDF Template



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Outline

Foundations

Structure and Math

Tables and Figures

Before You Share

Why This Template Exists

Slides often look polished while hiding their structure from assistive technology. This template compiles with LuaLaTeX to a tagged PDF/UA-2 file that exposes the same organization to a screen reader that a sighted viewer sees on screen.

Key idea

An accessible slide deck exposes real headings, real lists, real tables, and real math, not just visual layout that looks like them.

A Note on Overlays

This template avoids `\pause` and other overlay reveals on purpose.

- Each overlay step becomes a separate output page.
- A screen reader then encounters the same slide several times in a row, with only small differences.
- Every frame here is written as one complete, final page instead.

If you do use overlays

Check the tagged reading order afterward: duplicated near-identical pages are confusing without sight.

Headings Create Real Navigation

Every frame title in this deck is a real `\frametitle`, and every `\section` command both starts a PDF bookmark and groups related frames.

- Bookmarks let a reader jump between topics without scrolling.
- Tags let a screen reader announce a heading before a frame title, the same way it would for a printed article section.
- Neither happens automatically from font size or boldface alone.

Current limitation: `ltx-talk` tags frame titles at heading level H4 rather than H1, since the relationship between frame titles and document sectioning is still being decided upstream. Some checkers may flag this.

Displayed Equations

Use math environments rather than raw display delimiters, so the tagging layer can attach a MathML-oriented alternative to each formula.

$$J(\theta) = \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x_i) - y_i)^2 \quad (1)$$

$$\theta_{k+1} = \theta_k - \alpha \nabla J(\theta_k) \quad (2)$$

Lists Near Equations

A short list is often clearer than a dense paragraph when a formula has several symbols.

- m is the number of observations.
- $h_{\theta}(x_i)$ is the model prediction for observation i .
- y_i is the measured response.
- $\nabla J(\theta_k)$ is the gradient evaluated at iteration k .

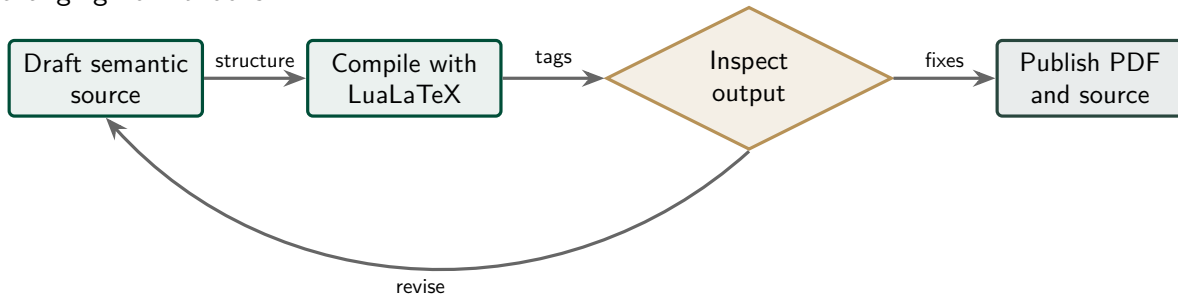
A Tagged Table

Table 1: Accessibility checks for a presentation PDF.

Check	Why it matters
Real frame titles	Screen readers and PDF outlines rely on headings.
Figure alt text	Visual information needs a text alternative.
Declared table header	Data cells need a header to remain meaningful out of order.
Equation environments	Math needs semantic structure, not just glyphs.

Tagged Vector Figure: Workflow

The `alt` key on the `tikzpicture` environment gives the diagram a text alternative without changing how it looks.



Writing Useful Alternative Text

Better

Contour plot of a bowl-shaped objective with a descent path moving toward the minimum.

Good alternative text names the figure type, states the main pattern or conclusion, and includes labels or trends the argument depends on. It rarely needs to list every visual detail.

Weaker

Image of a plot with green and gold lines.

Validation Checklist

Before Compiling

- Real frame titles and sections
- Equation environments, not raw $\$$
- Alt text for every meaningful figure
- Header row declared for every table

After Compiling

- Compiled with `lualatex`, twice
- PDF reports as tagged
- Title, author, and language are set
- Reading order matches the visual order

References

- World Wide Web Consortium. *Web Content Accessibility Guidelines (WCAG) 2.1*. W3C Recommendation, 2018.
- International Organization for Standardization. *ISO 14289-2:2024, Document management applications – Electronic document file format enhancement for accessibility – Part 2: Use of ISO 32000-2 (PDF/UA-2)*. ISO, 2024.

Questions?

Source: WM_Accessible_LuaLaTeX_Presentation.tex